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2.7 to 2.9% by mass. Approximately 50 lakh tons of salt are produced annually in India by solar evaporation. Crude sodium chloride can be obtained by crystallisation of brine solution which contains sodium sulphate, calcium sulphate, calcium chloride and magnesium chloride as impurities. Pure sodium chloride can be obtained from crude salt as follows. Firstly removal of insoluble impurities by filtration from the crude salt solution with minimum amount of water. Sodium chloride can be crystallised by passing HCl gas into this solution. Calcium and magnesium chloride, being more soluble than sodium chloride, remain in solution.

Sodium chloride melts at 1081K. It has a solubility of 36.0 g in 100 g of water at 273 K. The solubility does not increase appreciably with increase in temperature.

Uses :

(i) It is used as a common salt or table salt for domestic purpose.

(ii) It is used for the preparation of many inorganic compounds such as NaOH and Na_2CO_3

Sodium hydroxide:

Sodium hydroxide is prepared commercially by the electrolysis of brine solution in Castner-Kellner cell using a mercury cathode and a carbon anode. Sodium metal is discharged at the cathode and combines with mercury to form sodium amalgam. Chlorine gas is evolved at the anode. The sodium amalgam thus obtained is treated with water to give sodium hydroxide.

At cathode : $Na^+ + e^- \rightarrow Na(amalgam)$

At anode : $Cl^- \rightarrow \frac{1}{2} Cl_2 \uparrow + e^-$

$2Na(amalgam) + 2H_2O \rightarrow 2NaOH + 2Hg + H_2\uparrow$

Sodium hydroxide is a white, translucent and deliquescent solid, that dissolves in water to give a strong alkaline solution. It melts at 591 K. The sodium hydroxide solution at the surface reacts with the CO_2 in the atmosphere to form Na_2CO_3

Uses:

- Sodium hydroxide is used as a laboratory reagent
- It is also used in the purification of bauxite and petroleum refining
- It is used in the textile industries for mercerising cotton fabrics
- It is used in the manufacture of soap, paper, artificial silk and a number of chemicals

Sodium bicarbonate $NaHCO_3$ (Backing soda):

Sodium hydrogen carbonate or sodium bicarbonate is used in backing cakes pastries etc. It is called so because it decomposes on heating to generate bubbles of carbon dioxide, leaving holes in cakes or pastries and making them light and fluffy. This compound is prepared by saturating a solution of sodium carbonate with carbon dioxide. The white crystalline powder of sodium bicarbonate, being less soluble, precipitated out.